

Yan Li

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Research Interests

Trustworthy Medical AI; Multimodal AI for Healthcare; Medical Vision-Language Models; Clinical AI Agents.

Education

Beijing University of Posts and Telecommunications

Sept. 2025–present | Beijing, China

M.S. in Management Science and Engineering

Advisor: [Yawen Li](#); Co-advisor: [Guanhua Ye](#).

University of Electronic Science and Technology of China

Sept. 2021–July 2025 | Chengdu, China

Dual B.S. in Finance and Electronic Information Engineering

Thesis Advisor: [Bo Liu](#).

Research Experience

CAPRA: Multimodal Auditing under Missing Clinical Metadata

ACM MM 2026 Oral

Lead Student Researcher, Beijing University of Posts and Telecommunications

- Led CAPRA, a semantic-visual multimodal framework that fuses medical images with proxy-axis tokens and calibrated posteriors for missing-metadata subgroup auditing.
- Validated fundus, dermoscopy, and chest-radiography tasks under distribution shift, improving worst-group accuracy in 14 of 15 learner-dataset settings.

Selective-Dependence Auditing for Medical LLM Fairness

Under Review

Lead Student Researcher, Beijing University of Posts and Telecommunications

- Led the Selective-Dependence Audit, separating medical judgments, care pathways, support actions, and interface effects to distinguish prohibited dependence from appropriate clinical adaptation.
- Built ClinSDA-400 and evaluated six LLMs across 1,075,020 responses in semantic-label, JSON, and MCQ formats, isolating audit displacement caused by option order and answer-symbol mapping.

MedFuse: Anatomical-Prior Fusion for Retinal Lesion Segmentation

FCS 2026

Lead Student Researcher, Beijing University of Posts and Telecommunications

- Led technical development with an ophthalmologist at Peking University First Hospital, integrating RGB fundus features with retinal vascular priors for diabetic-retinopathy lesion segmentation.
- Evaluated a hierarchical dual-encoder fusion pipeline on DDR and IDRiD across five backbones, including annotation-free priors from a multimodal large language model.

Peer-Reviewed Publications

ACM MM 2026 ORAL Beyond Metadata: CAPRA for Hidden Subgroup Analysis under Missing Metadata in Medical Imaging

Yawen Li, **Yan Li**, Zhe Xue, Yingxia Shao, Meiyu Liang, and Guanhua Ye^{*}.

Proceedings of the 34th ACM International Conference on Multimedia, 2026.

ACM MM 2026 What Does Your Short-Answer VQA Score Actually Measure? Evaluator-Dependent Instability in Multimodal Short-Answer Benchmarks

Guanhua Ye, Jingbin Niu, **Yan Li**, Meiyu Liang, Zhe Xue, Yingxia Shao, and Yawen Li^{*}.

Proceedings of the 34th ACM International Conference on Multimedia, 2026.

AAAI 2026 Rethink Representation Learning for Questionnaire Data

Guanhua Ye, Jifeng He, **Yan Li**, Junping Du, Zhe Xue, Yingxia Shao, Meiyu Liang, and Yawen Li^{*}.

Proceedings of the AAAI Conference on Artificial Intelligence, 40(33):27782–27790, 2026.

TKDE 2025 FeBT: A Feature Balancing Transformer for Corporate ESG Forecasting

Yawen Li, Mengyu Zhuang, Guanhua Ye^{*}, **Yan Li**, Junheng Wang, Jinyi Zhou, and Pengfei Zhang.

IEEE Transactions on Knowledge and Data Engineering, 2025.

FCS 2026 MedFuse: A Multi-source Data Fusion Framework for Diabetic Retinopathy Lesion Segmentation

Jiantong Du, **Yan Li**, Yawen Li, Liwen Liao, Zhihao Zhao, and Guanhua Ye^{*}.
Frontiers of Computer Science, 20(3):2003625, 2026.

Conference Presentations

“Trust-free Personalized Decentralized Learning”	Feb. 2026
<i>Oral Presentation at IEEE BigComp 2026, Guangzhou, China.</i>	Paper Slides
“Slot-Based Evidence Routing for Budgeted Multi-Agent Coordination”	May 2026
<i>Oral Presentation at CINT 2026, Hangzhou, China.</i>	Paper Slides
“Auditing Historical Clinical Queries with Missing Availability Time”	Aug. 2026
<i>Oral Presentation at CCF BigData 2026, Harbin, China.</i>	Paper Slides

Teaching

Beijing University of Posts and Telecommunications

Big Data and Intelligent Management	Fall 2025
<i>Teaching Assistant, graduate course, School of Economics and Management</i>	
Operating Systems	Spring 2026
<i>Teaching Assistant, undergraduate course, School of Computer Science</i>	

Honors and Awards

University of Electronic Science and Technology of China

Certificate for Honors Research	2025
“Academic Star” Award of School of Management and Economics (sole recipient)	2024
Outstanding Student Scholarship	2024
Model Student Scholarship	2023